

A Coefficient-Space Extension of Matrix Hypercontractivity

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Source Question

Arunachalam and Doriguello ask in Future work item 4 of arXiv:2109.02600 whether one can prove a general (q, p) -hypercontractive statement for matrices, first over $\{0, 1\}^n$ and then over $\mathbb{Z}_r^n$. The relevant source passage is reproduced in Figure 1.

3. Better bounds on streaming algorithms. What is the quantum space complexity of approximating Unique Games? Given the recent work of Kallaugher and Parekh [45], who proved a $\Omega(n)$ streaming lower bound for Max- t -Cut, we conjecture that the same bound applies to Unique Games. A proof might require obtaining new quantum lower bounds for the communication problems introduced in [19, 20, 47, 48].

4. Generalized hypercontractivity. Another open question is regarding our main Result 1, which shows a form of $(2, p)$ -hypercontractivity, since the result works for all Schatten p -norms with $p \in [1, 2]$. Can we prove a general (q, p) -hypercontractive statement for matrices, firstly for matrix-valued functions over $\{0, 1\}$, and then further generalize that to functions over $\mathbb{Z}_r$? Proving this might also require a different generalization of the inequality of Ball, Carlen, and Lieb [5].

2 Preliminaries

Let $[n] \triangleq \{1, \dots, n\}$. For $r \in \mathbb{Z}$, $r \geq 2$, we let $\mathbb{Z}_r \triangleq \{0, \dots, r-1\}$ be the ring with addition and multiplication modulo r , and let $\omega_r \triangleq e^{2\pi i/r}$. Given $S \in \mathbb{Z}_r^n$, we write $|S| \triangleq |\{i \in [n] : S_i \neq 0\}|$ for its Hamming weight. Let $D(\mathbb{C}^m)$ be the set of all quantum states over $\mathbb{C}^m$, i.e., the set of positive semi-definite matrices with trace 1. For a matrix $M \in \mathbb{C}^{m \times m}$, its (unnormalized) Schatten p -norm is defined as $\|M\|_p \triangleq (\text{Tr } |M|^p)^{1/p} = (\sum_{i=1}^m \sigma_i(M)^p)^{1/p}$, where $\{\sigma_i(M)\}_i$ are the singular values of M , i.e., the eigenvalues of the positive semi-definite operator $|M| \triangleq \sqrt{M^\dagger M}$. Moreover, let $\|M\| \triangleq$

Figure 1: Future work item 4 in arXiv:2109.02600, PDF page 9.

This packet proves a partial answer for the Fourier-coefficient formulation suggested by their Result 1 / Theorem 3.3. It does not settle the broader physical-space formulation

$$\|T_\rho f\|_{L_q(S_q)} \leq \|f\|_{L_p(S_p)}.$$

Notation

Let $G = \mathbb{Z}_r^n$ with normalized counting measure and dual characters $\chi_S(x) = \omega_r^{S \cdot x}$, where $\omega_r = e^{2\pi i/r}$. For $f : G \rightarrow \mathbb{C}^{m \times m}$ set

$$\widehat{f}(S) = \mathbb{E}_{x \in G} f(x) \overline{\chi_S(x)}.$$

For $1 \leq p < \infty$ write

$$\|f\|_{L_p(S_p)} = (\mathbb{E}_{x \in G} \|f(x)\|_p^p)^{1/p}, \quad \|a\|_{\ell_q(S_p)} = \left(\sum_S \|a(S)\|_p^q \right)^{1/q}.$$

The usual modifications are made for $q = \infty$. Here $\|\cdot\|_p$ denotes the unnormalized Schatten p -norm.

For $1 \leq p \leq 2$ define

$$\beta_r(p)^2 = \frac{(p-1)(1-(p-1)^{r-1})}{(r-1)(2-p)},$$

with the continuous value $\beta_r(2) = 1$. This is the noise constant in Theorem 3.3 of arXiv:2109.02600.

Proof Intuition

The source theorem is already the endpoint $q = 2$ for the coefficient transform $f \mapsto \{\rho^{|S|} \widehat{f}(S)\}_S$. A second endpoint is free: Schatten-valued Hausdorff–Young sends $L_p(S_p)$ into $\ell_{p'}(S_p)$ without any noise loss. Interpolating the numerical sequence of coefficient norms between these two endpoints gives every $2 \leq q \leq p'$, and $\ell_q \leq \ell_{p'}$ gives the remaining $q \geq p'$ range. Finally, scalar degree-one character sums show that no dimension-free coefficient inequality can hold for $q < 2$.

Theorem

Theorem 1 (Coefficient (q, p) matrix hypercontractivity). *Let $r \geq 2$, $n, m \geq 1$, $1 \leq p \leq 2$, and $2 \leq q \leq \infty$. Let $p' = p/(p-1)$, with $p' = \infty$ when $p = 1$.*

If $2 \leq q < p'$, define $a = a(p, q)$ by

$$\frac{1}{q} = \frac{a}{2} + \frac{1-a}{p'}.$$

Then, for every $f : \mathbb{Z}_r^n \rightarrow \mathbb{C}^{m \times m}$ and every $0 \leq \rho \leq \beta_r(p)^a$,

$$\left(\sum_{S \in \mathbb{Z}_r^n} \rho^{q|S|} \|\widehat{f}(S)\|_p^q \right)^{1/q} \leq (\mathbb{E}_{x \in \mathbb{Z}_r^n} \|f(x)\|_p^p)^{1/p},$$

with the evident supremum norm interpretation when $q = \infty$.

If $q \geq p'$, including the endpoint $q = p'$, the same estimate holds for every $0 \leq \rho \leq 1$. For $p = 1$ this means that the finite- q conclusion has only the trivial $\rho = 0$ endpoint, while the $q = \infty$ endpoint has $\rho \leq 1$.

Proof. Put

$$A = \|f\|_{L_p(S_p)}, \quad b_S = \|\widehat{f}(S)\|_p.$$

The source theorem gives the coefficient endpoint

$$\left(\sum_S \beta_r(p)^{2|S|} b_S^2 \right)^{1/2} \leq A. \tag{1}$$

We also need the Schatten-valued Hausdorff–Young endpoint

$$\left(\sum_S b_S^{p'} \right)^{1/p'} \leq A, \tag{2}$$

with the usual supremum interpretation for $p = 1$. For completeness we recall the short proof. The Fourier transform is contractive from $L_1(S_1)$ to $\ell_\infty(S_1)$ by the triangle inequality. It is an isometry

from $L_2(S_2)$ to $\ell_2(S_2)$ by entrywise Parseval for the Hilbert–Schmidt norm. Finite-dimensional complex interpolation of Schatten classes and sequence spaces gives (2) for every $1 \leq p \leq 2$.

Assume first that $2 \leq q < p'$ and choose $a \in (0, 1]$ as in the statement. If $\rho \leq \beta_r(p)^a$, then

$$\begin{aligned} \sum_S \rho^{q|S|} b_S^q &\leq \sum_S \beta_r(p)^{aq|S|} b_S^q \\ &= \sum_S (\beta_r(p)^{|S|} b_S)^{aq} b_S^{(1-a)q}. \end{aligned}$$

Since $aq/2 + (1-a)q/p' = 1$, Holder’s inequality and (1)–(2) imply

$$\sum_S \rho^{q|S|} b_S^q \leq \left(\sum_S \beta_r(p)^{2|S|} b_S^2 \right)^{aq/2} \left(\sum_S b_S^{p'} \right)^{(1-a)q/p'} \leq A^q.$$

Taking q th roots proves the first range.

If $q \geq p'$, then $0 \leq \rho \leq 1$ and monotonicity of sequence norms give

$$\|\{\rho^{|S|} \widehat{f}(S)\}\|_{\ell_q(S_p)} \leq \|\{\widehat{f}(S)\}\|_{\ell_q(S_p)} \leq \|\{\widehat{f}(S)\}\|_{\ell_{p'}(S_p)} \leq A,$$

again using (2). This proves the theorem. $\square$

A Necessary Range Obstruction

Proposition 1. *Fix $1 \leq p \leq 2$ and $1 \leq q < 2$. There is no $\rho > 0$ and no constant $C < \infty$, independent of n , such that the coefficient estimate*

$$\left(\sum_S \rho^{q|S|} |\widehat{f}(S)|^q \right)^{1/q} \leq C (\mathbb{E}_x |f(x)|^p)^{1/p}$$

holds for all scalar functions on $\mathbb{Z}_2^n$.

Proof. Let $f_n(x) = \sum_{j=1}^n (-1)^{x_j}$ on $\mathbb{Z}_2^n$. Its only nonzero Fourier coefficients are the n degree-one coefficients, each equal to 1. Therefore the left side is $\rho n^{1/q}$. Since $p \leq 2$ and the measure is normalized,

$$(\mathbb{E} |f_n|^p)^{1/p} \leq (\mathbb{E} |f_n|^2)^{1/2} = \sqrt{n}.$$

The claimed inequality would force $\rho n^{1/q} \leq C\sqrt{n}$ for every n , impossible because $1/q > 1/2$. $\square$

Scope and Verification Notes

Theorem 1 gives a dimension-free general (q, p) theorem for the Fourier-coefficient version of matrix hypercontractivity, and Proposition 1 shows that the qualitative range $q \geq 2$ is best possible for that formulation.

This remains a partial answer to the source future-work question, because the source question is informal and may also intend a physical-space $L_p(S_p) \rightarrow L_q(S_q)$ estimate for the noise operator. No such estimate is proved here, and no optimality of the noise constants in Theorem 1 is claimed.

The duplicate/literature check searched the run indexes for arXiv:2109.02600, the title, matrix hypercontractivity, generalized (q, p) hypercontractivity, Schatten, and $\mathbb{Z}_r$, and also searched the web for exact combinations of “matrix-valued hypercontractivity”, “q,p”, “Schatten”, and “Z_r”. No existing run packet or exact later answer to this coefficient formulation was found.

References

1. S. Arunachalam and J. F. Doriguello, *Matrix hypercontractivity, streaming algorithms and LDCs: the large alphabet case*, arXiv:2109.02600; ACM Transactions on Computation Theory 16(4), Article 21, 2024.